Supplementary Online Content

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This supplementary material has been provided by the authors to give readers additional information about their work.

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eMethods

Supplemental Methods

Detailed methodology such as full inclusion/exclusion criteria are published (clinicaltrials.gov NCT04995029).[1]

Screening

At screening, participants were assessed for eligibility including a detailed opioid use history in the past 28 days. Participants who met the criteria for high-risk opioid use were those who used opioids for an average of ≥5 days per week in the last 4 weeks and used opioids via the injection route, used high doses of opioids, or used fentanyl/fentanyl analog**s** by any route.

Induction to open-label treatment

A nested open-label induction study randomized participants to standard (≥7-day-period of transmucosal buprenorphine, BUP-TM, dose adjustments) vs. rapid (with a single dose of BUP-TM) induction onto extended release buprenorphine, BUP-XR.[2] Participants received 2 injections of 300-mg BUP-XR spaced 1 week apart before being randomized.

Randomization to double-blind treatment

Four weeks after the second open-label BUP-XR injection, participants were randomized to receive 8 additional maintenance doses of either 100 mg or 300 mg every 4 weeks (Figure 1). Randomization schedules were generated using a balanced-across-centers approach,[3] with block size of 4 in a 1:1 ratio. Four distinct blocks (CDCD, CDDC, DCDC, DCCD) formed a balanced set for Latin Squares, then the block order was randomly permuted to form a sequence of Latin Squares. A block was dynamically allocated to a site by interactive web response system (IWRS), at the time of randomizing the first participant at that site. This approach helps balance the treatment assignments across sites when blocks are left incomplete. Note, blocks CCDD and DDCC were excluded to reduce the probability of imbalance within a site.

Blinding

The blinding of the study was maintained as follows:

- An Interactive Web Response System (IWRS) was used to track the randomization and to assign the treatment for each study participant and it was responsible for subsequent assignment of all BUP-XR doses
- Each study site had an unblinded study drug manager and an unblinded drug administrator who had no other responsibilities in the clinical trial (fulfilled by the same or multiple individuals). The person who administered the treatment had to be a healthcare provider familiar with giving injections
- At each injection visit during double-blind treatment (injections 3–10), an unblinded staff member contacted the IWRS to obtain the treatment assignment
- The unblinded staff member then obtained and prepared the BUP-XR for administration. Because of volume and size differences between BUP-XR containing 100 mg or 300 mg. The unblinded staff member applied a large blank label around the barrel of the syringe several times to mask the volume and size of the injection. BUP-XR preparation occurred

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in a separate room away from blinded study staff and the participant. Study participants and blinded site personnel were shielded from seeing the syringe prior to and during the injection

- Once BUP-XR was prepared, the unblinded study drug administrator administered BUP-XR to the participant as described in the applicable product labelling information and this was tracked via IWRS and drug accountability logs by the unblinded study drug manager for each participant

Double-blind BUP-XR maintenance treatment and assessment of drug use

From weeks 1–10, participants returned to the site for weekly urine drug screen (UDS) and collection of self-reported drug use (including timeline follow-back, TLFB).[4,5] From the fourth BUP-XR injection until the end of the treatment period (weeks 10–38), UDS and self-reported drug use (including TLFB) was obtained at every injection visit prior to receiving BUP-XR. In addition, random visits between injections (2 weeks post each injection ±7 days) were scheduled by the investigator to complete UDS and self-report only.

The UDS tested for opioids (including oxycodone, fentanyl, methadone, and morphine) and for other drugs including cocaine, amphetamines, methamphetamine, ecstasy, cannabinoids, barbiturates, benzodiazepines, and phencyclidine. UDS fentanyl data reflected only urine fentanyl-positive samples, except in the post hoc analysis of the primary endpoint that also included norfentanyl when deriving a positive or negative UDS result.

The TLFB asked participants to retrospectively report their daily drug use for each of the past 7 days prior to the visit. This captured if any opioid was used for a specific day ("used" or "did not use") specific opioid drugs (oxycodone, fentanyl, morphine, methadone, buprenorphine), or by drug class for other substances (cocaine, cannabinoids, barbiturates, benzodiazepines, amphetamines/methamphetamine, and phencyclidine). Additional self-report questions included the main route of opioid use (and more specifically, if an injectable route was used) and the number of times opioids were used each day.

All participants received counselling, per standard of care, from the first induction day through the end of the treatment visit (week 38/EOT), four weeks after injection 9. Participants who prematurely discontinued BUP-XR treatment completed the early termination visit (ET). During the last injection visit (week 34) through the EOT visit (or ET for those who prematurely discontinued), the investigator or a medically qualified sub-investigator discussed available options for continued treatment. Any participant with ongoing adverse events (AEs) at the EOT or ET visit was followed up by phone 2 weeks later for the end of study visit to assess any ongoing AEs and concomitant medications associated with those ongoing AEs only.

Participants were not permitted to use supplemental BUP-TM after they left the site on day 1. No dose change of BUP-XR were allowed. If the participant experienced withdrawal symptoms at any time, they could be treated symptomatically.[6]

Statistical Analysis

<u>Sample size</u>

The assumptions for sample size calculation necessary was based on post-hoc analysis of participants using opioids via the injection route in prior phase 3 clinical results. These results showed that the proportion of responders (where a responder was defined as a participant having

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≥ 80% of weeks with opioid use abstinence [UDS and self-report combined] during weeks 10–25 corresponding to assessments post BUP-XR Injections 3–6) were 44.4% and 28.8% for 300-mg and 100-mg maintenance doses, respectively, leading to a responder rate difference (95% confidence interval, CI) of 15.66% (–0.77%, 32.08%). Thus, for the primary efficacy endpoint (proportion of responders with ≥80% opioid abstinence over weeks 20–38), a sample size of 195 per group would provide approximately 90% power at 2-sided 0.05 alpha level to detect a difference of 15.6% (44.4% vs. 28.8%). Since the study population was more severe in this study compared to the prior phase 3 double-blind study, the responder rates and treatment differences may be lower. To evaluate this possibility, the responder rates required for the BUP-XR 300 mg group in order to achieve at least 80% power with N=195 per group under various assumptions for responder rates in the BUP-XR 100 mg group are summarized in **eTable 2**, as well as the power under various assumptions for the BUP-XR 100 mg and 300 mg groups and using 2-sided 0.05 alpha level with N=195 per group.

Derivation of opioid use combining UDS and self-reported opioid use for a visit

Firstly, the overall self-reported opioid use based on the 7-daily TLFB for the prior week at a visit was derived according to the following table:

Overall Self-Reported Opioid Use Derivation Based on Daily TLFB Results at a Given Visit

	Daily TimeLine Follow Back Opioids		
	All Days=Missing	**All Nonmissing Days=Did Not Use**	**Any Day=Used**
Overall Self-Reported Opioids	Missing	Negative	Positive

The opioid use at a visit combining UDS and overall self-reported results was derived according to following table:

Opioid Use Derivation Based on UDS and Overall Self-Reported Opioid Use Results at a Given Visit

		Overall Self-Reported Opioids		
		Missing	**Negative**	**Positive**
Urine Drug Screen Opioids	**Missing**	Missing	Negative	Positive
	Negative	Negative	Negative	Positive
	Positive	Positive	Positive	Positive

Sensitivity and exploratory analyses for the primary endpoint

Multiple imputation was used to address missing opioid assessments at individual visits for the primary endpoint. A standard, three-step approach was used:[7]

- Imputation Phase: 50 imputed datasets were generated separately in treatment group, using Fully Conditional Specification (FCS) Logistic Regression[7-9] to impute intermittent and monotone missing visits for opioid use status from randomization through Week 38 using SAS Proc MI. The imputation model included the following variables as covariates in addition to non-missing visit results: Sex, Race Group (Black vs Non-Black), Age, BMI, Lifetime Opioid Use (years), Percentage (%) of Abstinence During OL Injections,

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Baseline Nicotine use (Y/N), Baseline Alcohol use (Y/N), Baseline Fentanyl use (Y/N), Baseline Cocaine use (Y/N), Baseline Marijuana/Cannabinoids use (Y/N), Baseline Amphetamines/Methamphetamines use (Y/N), Baseline OC-VAS score (worst craving in the past week), Baseline Employment Status (Y/N), Baseline PROMIS Sleep Disturbance Score, Pre-existence of a Psychiatric disorder (Y/N), OUD Severity (Moderate vs Severe), prior detox from opioids (Y/N), and finally, prior experience of an overdose (Y/N).

- Analysis Phase: After imputation, the primary endpoint was re-derived using full visit data for each participant in each of the imputed datasets. The 50 datasets were then each analyzed separately, by imputation, to obtain the Cochran-Mantel-Haenszel (CMH) weighted treatment difference and its associated variance estimation, as was produced with the primary endpoint
- Pooling Phase: Resulting estimates from the analysis phase was combined using SAS Proc MIANALYZE (SAS EG Version 7.1, SAS Institute Inc., Cary, North Carolina) to produce combined estimates for treatment difference (95% CI) and associated P-value.

As an additional exploratory analysis, the primary endpoint was analyzed via Bayesian approach (using SAS Proc MCMC). The posterior distribution of the proportion of responders for each treatment arm was estimated via a Bayesian beta-binomial model. The model assumed a non-informative, uniform, conjugate Beta (α, β) prior for the responder probability of each treatment arm (i.e., Beta [1,1]). The difference of responder proportion (95% highest posterior density credible interval) and posterior probability of 300 mg superiority over 100 mg was also estimated.

Secondary endpoints

The key secondary endpoint:

Participants' percentage of days with opioid abstinence out of days assessed (weeks 10–38 inclusive) was based on the 7-daily self-report assessments for the prior week. Percentage for an individual participant was derived as 1 minus their number of days of opioid use divided by the number of days with observed self-report information. Missing daily self-report data (monotone or intermittent) were not imputed. Abstinence for participants who did not reach week 10 was imputed based on the self-reported use in weeks 7–9 or week 6 if a participant discontinued right after DB randomization. A Wilcoxon rank sum test[10] stratified for randomization factors was performed to compare the difference between the 2 treatment arms.

Other secondary endpoints:

The average number of times opioids were used per week for a given visit of a participant was based on the daily self-report for the prior week collected at that visit and calculated as the total number of times opioids were used in a week divided by the number of days with observed self-report information for that visit, then multiplied by 7. The average value for each treatment arm was summarized by visit.

The average number of days that opioids were used per week for a given visit of a participant was calculated as the number of days opioids were used in a week divided by the number of days with observed self-report information for that visit, then multiplied by 7.

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Post-hoc subgroup analyses

To explore the association between treatment response and characteristic variables collected at screening (re-treatment), multivariable regressions were performed with outcome variable of either participant's percentage of visits with abstinence over weeks 20–38 (linear regression) or the primary endpoint (responder), defined as abstinence ≥80% from weeks 20–38 (logistic regression).

The baseline factors included in the full model consisted of the following: double-blind randomization arm (300 mg/100 mg), sex (female/male), race group (Black or African American/Non-Black or African American), the actual open-label randomized induction arm (standard/rapid induction, SI/RI; participants in an "alternate arm" were assigned to the SI group for analysis purposes), screening BMI, the number of times opioids were reported as being used per day at the screening self-report, the minimum number of times fentanyl was reported as being used per week at the screening self-report, daily fentanyl use (yes/no) at screening, the number of additional drug classes used (ordinal), and the main route of opioid use as recorded in drug use history. While the number of times fentanyl was used per week was not collected by direct question, the minimum number of times fentanyl used the week before screening was estimated as the total number of times opioids were used on days where participants reported using fentanyl only, +1 for each day where fentanyl use was reported in combination with other opioids. This number is regarded as a minimum estimate because the participants could have used fentanyl more than once per day on any of the days for which they answered positively for fentanyl in combination with other opioids.

The backward selection algorithm with *P*-value=0.02 was used to identify a reduced set of variables that is correlated with the outcome (reduced model). In this process, an explanatory variable with largest *P*-value (>0.02) was removed from the full model until all the remained variables had *P*-value ≤0.2.

(1) Linear multivariable regression with outcome variable of participants' percentage of visits with abstinence over weeks 20–38 as the outcome (ranging 0–100). The reduced model (**eTable 3)** retained the Baseline factors of sex, race group, number of times opioids were used per day at Screening, and daily fentanyl use at Screening. Using fentanyl daily significantly associated with lower abstinence and was the only factor with a *P*-value <0.05 (*P*=0.0365). After maintenance dosage group and associated interactions with the retained factors were placed back into the reduced model, a backward selection (*P*-value=0.02) was performed again, with main effects and 'dosage group' being forced in the model. The interaction term of daily fentanyl use (*P*=0.0447) was retained (**eTable 4**), indicating the participants who use fentanyl daily had 15% higher percentage of abstinence in the 300-mg group than those in the 100-mg group.

(2) Logistic multivariable regression using primary endpoint as outcome (responder yes vs. no). The baseline factors of sex, BMI, minimum number of times fentanyl used the week prior to screening, and the main route of opioid use were retained in the reduced model (**eTable 5**) with the minimum number of times fentanyl was used per week at screening being the only factor with a *P*-value <0.05 (*P*=0.0103). Participants who reported using fentanyl more often had a significantly lower likelihood of becoming a responder. A cut-point value was then estimated for the continuous main effect variable in the reduced model, via receiver operating characteristic (ROC) analysis. The value on the ROC curve associated with the maximum

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Youden's J (sensitivity+specificity-1) was selected as the optimal threshold, to dichotomize the continuous main effect variable for the assessment of potential interaction with the maintenance dosage group. The value of 14 was estimated as the optimal cut-points for the minimum number of times fentanyl used the week prior to screening, and 28 was estimated for BMI. Replacing continuous variables with dichotomized variables in the reduced model slightly increased the C-statistics (area under the ROC curve) from 0.635 to 0.656. A similar backward selection for the interaction terms between the maintenance dosage group and the main effect variables in the reduced model was performed, as described for (1). A significant interaction term was identified only between 300-mg maintenance dose and the participants who used fentanyl a minimum of ≥14 times the week before Screening (P=0.0399), suggesting a higher likelihood of becoming a responder for the participants who used fentanyl a minimum of ≥14 times the week before Screening when receiving 300 mg (**eTable 6**).

Finally, the primary endpoint was reassessed again for the high-frequency fentanyl using subgroups newly identified via the above 2 criteria: Daily fentanyl use or using fentanyl a minimum of ≥14 times the week before Screening.

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eTables

eTable 1. Treatment Emergent Adverse Events[a]

	Maintenance Dosage Arm		Total (N = 435) n (%)	Difference (95% CI) (300 mg – 100 mg)[b]
	100 mg (N = 218) n (%)	300 mg (N = 217) n (%)		
BUP-XR initiation (starting from injection 1 and up to injection 3)				
Any TEAE on or after the date of the 1st injection and prior to injection 3[c]	86 (39.45)	106 (48.85)	192 (44.14)	9.40 (0.06, 18.58)
BUP-XR–related TEAEs	51 (23.39)	73 (33.64)	124 (28.51)	10.25 (1.76, 18.63)
Severe TEAEs	5 (2.29)	6 (2.76)	11 (2.53)	0.47 (-2.85, 3.90)
Serious TEAEs	2 (0.92)	0	2 (0.46)	-0.92 (-3.29, 0.83)
Serious TEAEs meeting liver criteria[4]	0	0	0	NA
Fatal TEAEs	0	0	0	NA
Hepatic disorder TEAEs (CMQ)[e]	3 (1.38)	1 (0.46)	4 (0.92)	-0.92 (-3.56, 1.31)
Injection site reaction TEAEs (CMQ)[e]	15 (6.88)	19 (8.76)	34 (7.82)	1.88 (-3.30, 7.17)
BUP-XR –related TEAEs excluding injection site reaction TEAEs (CMQ)[e]	41 (18.81)	63 (29.03)	104 (23.91)	10.22 (2.22, 18.19)
Opioid withdrawal symptom TEAEs	31 (14.22)	45 (20.74)	76 (17.47)	6.52 (-0.64, 13.72)
TEAEs leading to BUP-XR discontinuation	0	0	0	NA
TEAEs of Special Interest[f]	0	0	0	NA
Double blind treatment period (starting from injection 3)				
Any TEAE on or after the date of the 1st DB maintenance injection/ injection 3[c]	100 (45.87)	116 (53.46)	216 (49.66)	7.58 (-1.82, 16.86)
BUP-XR–related TEAEs	22 (10.09)	44 (20.28)	66 (15.17)	10.18 (3.50, 17.01)
Severe TEAEs	9 (4.13)	13 (5.99)	22 (5.06)	1.86 (-2.42, 6.35)
Serious TEAEs	11 (5.05)	11 (5.07)	22 (5.06)	0.02 (-4.36, 4.42)
Serious TEAEs meeting liver criteria[d]	0 (0.0)	0 (0.0)	0 (0.0)	NA
Fatal TEAEs	0	1 (0.46)	1 (0.23)	0.46 (-1.28, 2.57)
Hepatic disorder TEAEs (CMQ)[e]	6 (2.75)	6 (2.76)	12 (2.76)	0.01 (-3.46, 3.50)
Injection site reaction TEAEs (CMQ)[e]	12 (5.50)	32 (14.75)	44 (10.11)	9.24 (3.70, 15.16)
BUP-XR–related TEAEs excluding injection site reaction TEAEs (CMQ)[e]	15 (6.88)	24 (11.06)	39 (8.97)	4.18 (-1.24, 9.80)
Opioid Withdrawal Symptom TEAEs	7 (3.21)	15 (6.91)	22 (5.06)	3.70 (-0.45, 8.27)
TEAEs leading to BUP-XR discontinuation	0	1 (0.46)	1 (0.23)	0.46 (-1.28, 2.57)
TEAEs of special interest[f]	0 (0.0)	0 (0.0)	0 (0.0)	NA

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	Maintenance Dosage Arm		Total	Difference (95% CI)
	100 mg (N = 218) n (%)	300 mg (N = 217) n (%)	(N = 435) n (%)	(300 mg – 100 mg)[b]
BUP-XR initiation (starting from injection 1 and up to injection 3)				

Shown for the safety population, defined as all randomized participants who received BUP-XR. A participant with more than 1 TEAE within a category was counted once for that category. The numerator is the number of participants with at least 1 TEAE. The denominator is the number of participants in each arm.
[a]Participants who received at least 1 BUP-XR maintenance injection post DB Randomization. The participants are analyzed corresponding to the maintenance dosage they actually received.
[b]Difference in the number of participants with an AE. Miettinen-Nurminen CI calculated without randomization strata.
[c]A DBTP TEAE was defined as an AE having an onset date/time on or after the date/time of Injection 3.
[d]Laboratory values of ALT or AST >3×ULN and bilirubin >2×ULN.
[e]Per the customized MedDRA Query (CMQ) definitions.
[f]TEAEs of special interest include Removal of Drug Depot.
BUP-XR extended-release buprenorphine; DB, double blind; AE, adverse event; CMQ, Customized MedDRA Query; LFT, liver function test; MedDRA, Medical Dictionary for Regulatory Activities; TEAE, treatment-emergent adverse event; ULN, upper limit of normal.

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eTable 2. Sample Size Evaluation Under Different Assumptions

BUP-XR 300/300 mg responder rate for the binary primary endpoint comparison required under assumptions of a lower BUP-XR 300/100 mg responder rate

BUP-XR 100 mg responder rate	Required BUP-XR 300 mg responder rate to achieve at least 80% power	Difference between 300 mg and 100 mg
5%	13.2%	8.20%
10%	20.2%	10.20%
15%	26.5%	11.50%
20%	32.5%	12.50%
25%	38.2%	13.20%
28.8%	42.3%	13.50%

Power under various assumed responder rates for BUP-XR 100 mg and 300 mg maintenance doses

	BUP-XR 300 mg responder rate						
BUP-XR 100 mg responder rate	**10%**	**15%**	**20%**	**25%**	**30%**	**35%**	**40%**
5%	47%	91%	99%				
10%		32%	79%	98%	99%		
15%			25%	70%	95%	99%	
20%				22%	63%	92%	99%
25%					20%	58%	89%
30%						18%	54%

Power calculations used 2-sided alpha=0.05 and 195 participants per group.
BUP-XR, extended-release buprenorphine.

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eTable 3. Multivariable Analysis for Participants' Percentage of Abstinence Week 20-38

Covariate	Full Model[a]		Reduced Model[b]	
	Estimate (95% CI)	Type III *P*-value	Estimate (95% CI)	Type III *P*-value
Observations used/read: n/n	434/434		434/434	
Double-blind randomization arm: 300 mg vs.100 mg	1.00 (-6.32, 8.32)	0.7884	NA	NA
Sex: Female/Male	5.92 (-1.48, 13.32)	0.1168	6.09 (-1.26, 13.43)	0.1041
Race: Black or African/Non-Black or African American	6.30 (-3.21, 15.81)	0.1939	6.76 (-2.22, 15.75)	0.1397
Actual open-label randomized induction arm[c]: SI/RI	4.17 (-3.50, 11.83)	0.2857	NA	NA
Screening BMI	-0.25 (-0.85, 0.34)	0.4063	NA	NA
Minimum number of times fentanyl used/week at screening[d]	-0.04 (-0.20, 0.12)	0.6007	NA	NA
Number of times opioids used/day at screening[e]	-0.54 (-1.47, 0.39)	0.2509	-0.64 (-1.34, 0.06)	0.0735
Daily fentanyl use at screening (7 days/week): Yes/No	-8.30 (-16.82, 0.22)	0.0561	-7.91 (-15.31, -0.50)	0.0365
Number of additional drug classes used (ordinal)[f]	-1.20 (-4.31, 1.92)	0.4504	NA	NA
Main route of opioid use at screening[g]		0.3233		NA
Smoking/injection	5.39 (-4.01, 14.78)		NA	
Snorting/injection	-1.06 (-11.19, 9.07)		NA	
Oral/injection	15.78 (-10.18, 41.74)		NA	

Shown for the efficacy population, defined as eligible participants who were randomized to double-blind treatment and received ≥1 BUP-XR maintenance injection and excluding 1 erroneously randomized participant who discontinued after BUP-XR injection 3 due to not meeting eligibility criteria. Opioid abstinence was defined as negative UDS and self-report for opioid use over Weeks 20 to 38 (inclusive, based on 6 scheduled UDS and self-report assessments and 4 planned random UDS and self-report assessments). Missing visit was counted as positive use.

The denominator shown in the ratio for dichotomous predictors was used as the reference group. Full population may not have been included in the model when covariates were missing.

[a]Based on a generalized linear model including all displayed covariates as predictors of participant abstinence % from Weeks 20-38.

[b]Based on a reduced generalized linear model predicting abstinence over Weeks 20-38 where covariates with a Type III p-value >0.20 were not retained in the model via a backwards stepwise selection process.

[c]Based on the actual induction arm. Alternative Induction was grouped with SI.

[d]Sum the number of times opioids were used on days where fentanyl ONLY was reported and counting 1 per day on days where Fentanyl use was reported in addition to other opioids in the past 7 days at Screening (self-report). This is an estimate of the minimum number of times fentanyl was used per week at Baseline.

[e]Calculated as the number times opioids were used on self-report days with reported opioid use out of the number of days with observed self-report information recorded at the Screening Visit.

[f]Drug classes include cocaine, marijuana/cannabinoids, barbiturates, benzodiazepines, amphetamines/methamphetamine, phencyclidine, LSD, and ecstasy as assessed at the Screening Baseline Visit. Phencyclidine, LSD, and ecstasy were combined into 1 category (hallucinogenic).

[g]"Injecting participants" was used as the reference group. Oral includes participants who checked "Other".

BUP-XR extended-release buprenorphine; CI, confidence interval; DB, double-blind; n, number of participants included in analysis; N, total participants; NA, not applicable; OL, open-label; RI, rapid induction; SI, standard induction; UDS, urine drug screen.

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eTable 4. Multivariable Analysis for Participants' Percentage of Abstinence Week 20-38: Interactions With Maintenance Dose

Covariate	Full Model[a]		Reduced Model[b]	
	Estimate (95% CI)	Type III P-value	Estimate (95% CI)	Type III P-value
Observations used/read: n/N	434/434		434/434	
Double-blind Randomization Arm: 300mg vs.100mg	-13.13 (-28.31, 2.05)	0.9419	-5.33 (-15.36, 4.70)	0.5699
Sex: Female/Male	6.49 (-3.88, 16.86)	0.0888	5.99 (-1.33, 13.32)	0.1086
Race: Black/Non-Black	1.46 (-11.28, 14.19)	0.1475	6.72 (-2.25, 15.68)	0.1417
Number of times opioids used/day at screening[c]	-1.25 (-2.38, -0.12)	0.0293	-0.71 (-1.41, -0.00)	0.0489
Daily fentanyl use at screening: y/n	-15.77 (-26.15, -5.38)	0.0313	-15.35 (-25.64, -5.06)	0.0358
Double-blind randomization arm x sex	-0.22 (-14.92, 14.48)	0.9766	NA	NA
Double-blind randomization arm x race	10.34 (-7.61, 28.29)	0.2583	NA	NA
Double-blind randomization arm x number of times opioids used/day at screening	0.90 (-0.55, 2.34)	0.2224	NA	NA
Double-blind randomization arm x daily fentanyl use at screening	15.23 (0.39, 30.07)	0.0442	14.84 (0.35, 29.33)	0.0447

Shown for the efficacy population, defined as eligible participants who were randomized to double-blind treatment and received ≥1 BUP-XR maintenance injection and excluding 1 erroneously randomized participant who discontinued after BUP-XR injection 3 due to not meeting eligibility criteria. A responder was defined as a participant whose percentage of visits with opioid abstinence (defined as negative UDS and self-report for opioid use) was ≥80% over Weeks 20 to 38 (inclusive, based on 6 scheduled UDS and self-report assessments and 4 planned random UDS and self-report assessments). Missing visit was counted as positive.

[a]Based on a Generalized Linear model including all displayed covariates as predictors of participant abstinence % from Weeks 20-38.

[b]Based on a Reduced Generalized Linear model predicting abstinence over weeks 20-38 where covariates with a Type III p-value >0.20 are not retained in the model via a backwards stepwise selection process.

[c]Calculated as the number times opioids were used on self-report days with reported opioid use out of the number of days with observed self-report information recorded at the screening visit.

The denominator shown in the ratio for dichotomous predictors was used as the reference group. Full population may not be included in the model when covariates are missing.

BUP-XR extended-release buprenorphine; CI, confidence interval; n, Number of participants included in analysis; N, total participants; OL, Open-label; UDS, urine drug screen.

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eTable 5. Logistic Analysis of the Primary Endpoint for Proportion of Responders With Weekly Opioid Use (Week 20-38)

Covariate	Full Model[a]			Reduced Model[b]		
	Estimate (95% CI)	Odds Ratio (95% CI)	Type III *P*-value	Estimate (95% CI)	Odds Ratio (95% CI)	Type III *P*-value
Observations used/read: n/N	434/434			434/434		
Double-blind randomization Arm: 300 mg vs. 100 mg	0.167 (-0.308, 0.641)	1.182 (0.735, 1.899)	0.4904	NA	NA	NA
Sex: Female/Male	0.473 (-0.004, 0.950)	1.605 (0.996, 2.587)	0.0521	0.439 (-0.030, 0.909)	1.552 (0.971, 2.481)	0.0665
Race: Black or African/non-Black or African-American	0.254 (-0.334, 0.843)	1.290 (0.716, 2.322)	0.3963	NA	NA	NA
Actual open-label randomized induction arm[c]: SI/RI	0.261 (-0.228, 0.751)	1.299 (0.796, 2.119)	0.2956	NA	NA	NA
Screening BMI	-0.037 (-0.078, 0.004)	0.964 (0.925, 1.004)	0.0780	-0.041 (-0.082, 0.000)	0.960 (0.921, 1.000)	0.0527
Minimum number of times fentanyl used/week at screening[d]	-0.011 (-0.026, 0.003)	0.989 (0.975, 1.003)	0.1210	-0.014 (-0.025, -0.003)	0.986 (0.976, 0.997)	0.0103
Number of times opioids used/day at screening[e]	-0.037 (-0.108, 0.035)	0.964 (0.898, 1.035)	0.3155	NA	NA	NA
Daily fentanyl use at screening (7 days/week): Yes/No	-0.005 (-0.601, 0.591)	0.995 (0.548, 1.806)	0.9874	NA	NA	NA
Number of additional drug classes used (ordinal)[f]	0.045 (-0.159, 0.250)	1.046 (0.853, 1.284)	0.6654	NA	NA	NA
Main route of opioid use at screening[g]			0.2300			0.1970
Smoking/injection	0.294 (-0.313, 0.901)	1.342 (0.731, 2.462)		0.216 (-0.376, 0.807)	1.240 (0.686, 2.242)	
Snorting/injection	0.078 (-0.578, 0.734)	1.081 (0.561, 2.082)		0.126 (-0.515, 0.767)	1.134 (0.597, 2.154)	
Oral/injection	1.397 (-0.038, 2.832)	4.044 (0.963, 16.978)		1.469 (0.125, 2.813)	4.346 (1.133, 16.666)	

Shown for the efficacy population, defined as eligible participants who were randomized to double-blind treatment and received ≥1 BUP-XR maintenance injection and excluding 1 erroneously randomized participant who discontinued after BUP-XR injection 3 due to not meeting eligibility criteria. A responder was defined as a participant whose percentage of visits with opioid abstinence (defined as negative UDS and self-report for opioid use) was ≥80% over Weeks 20 to 38 (inclusive, based on 6 scheduled UDS and self-report assessments and 4 planned random UDS and self-report assessments). The denominator shown in the ratio for dichotomous predictors was used as the reference group. Full population may not have been included in the model when covariates were missing.

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Covariate	Full Model[a]			Reduced Model[b]		
	Estimate (95% CI)	Odds Ratio (95% CI)	Type III *P*-value	Estimate (95% CI)	Odds Ratio (95% CI)	Type III *P*-value

[a]Based on a logistic model, including all displayed covariates as predictors of the primary endpoint response.
[b]Based on a reduced logistic model predicting the risk of primary endpoint response where covariates with a *P*-value >0.20 were not retained in the model via a backwards stepwise selection process.
[c]Based on the actual Induction arm. Alternative Induction was grouped with SI.
[d]Sum of the number of times opioids were used on days where fentanyl ONLY was reported and counting 1 per day on days where fentanyl use was reported in addition to other opioids in the past 7 days at Screening (self-report). This is an estimate of the minimum number of times fentanyl was used per week at Baseline.
[e]Calculated as the number times opioids were used on self-report days with reported opioid use out of the number of days with observed self-report information recorded at the Screening Visit.
[f]Drug classes include cocaine, marijuana/cannabinoids, barbiturates, benzodiazepines, amphetamines/methamphetamine, phencyclidine, LSD, and ecstasy as assessed at the Screening Baseline Visit. Phencyclidine, LSD, and ecstasy were combined into 1 category (hallucinogenic).
[g]"Injecting participants" was used as the reference group. Oral includes participants who checked "Other".
BUP-XR extended-release buprenorphine; CI, confidence interval; DB, double-blind; LSD, lysergic acid diethylamide; n, number of participants included in analysis; N, total participants; NA, not applicable; OL, open-label; RI, rapid induction; SI, standard induction; UDS, urine drug screen.

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eTable 6. Logistic Analysis of the Primary Endpoint for Proportion of Responders With Weekly Opioid Use (Week 20-38): Interactions With Maintenance Dose

Covariate	Full Model[a]		Reduced Model[b]	
	Estimate (95% CI)	Type III *P*-value	Estimate (95% CI)	Type III *P*-value
Observations used/read: n/N	434/434		434/434	
Double-blind randomization Arm: 300 mg vs.100 mg	0.030 (-0.998, 1.059)	0.9542	0.135 (-0.518, 0.789)	0.6850
Sex: Female/Male	0.249 (-0.442, 0.940)	0.4795	0.393 (-0.082, 0.868)	0.1049
Screening BMI ≥28	0.040 (-0.700, 0.780)	0.9152	0.023 (-0.709, 0.755)	0.9505
Minimum number of times fentanyl used ≥14/week at screening[c]	-1.637 (-2.546, -0.728)	0.0004	-1.576 (-2.460, -0.693)	0.0005
Main route of opioid use at screening[d]		0.4270		0.1749
Smoking/injection	0.529 (-0.311, 1.369)		0.288 (-0.312, 0.888)	
Snorting/injection	-0.076 (-1.050, 0.897)		0.230 (-0.428, 0.888)	
Oral/injection	1.116 (-1.732, 3.963)		1.554 (0.168, 2.939)	
Double-blind randomization arm x sex	0.277 (-0.678, 1.232)	0.5695	NA	NA
Double-blind randomization arm x BMI ≥28	-1.033 (-2.117, 0.051)	0.0618	-0.985 (-2.060, 0.090)	0.0724
Double-blind randomization arm x minimum number of times fentanyl used ≥14/week at screening[c]	1.233 (0.093, 2.373)	0.0340	1.142 (0.052, 2.231)	0.0399
Double-blind randomization arm x main route of opioid use at screening[d]		0.4237		NA
Smoking/injection	-0.459 (-1.661, 0.743)		NA	
Snorting/injection	0.545 (-0.786, 1.877)		NA	
Oral/injection	0.514 (-2.761, 3.789)		NA	

Shown for the efficacy population, defined as eligible participants who were randomized to double-blind treatment and received ≥1 BUP-XR maintenance injection and excluding 1 erroneously randomized participant who discontinued after BUP-XR injection 3 due to not meeting eligibility criteria. A responder was defined as a participant whose percentage of visits with opioid abstinence (defined as negative UDS and self-report for opioid use) was ≥80% over Weeks 20 to 38 (inclusive, based on 6 scheduled UDS and self-report assessments and 4 planned random UDS and self-report assessments).

[a]Based on a logistic model, including all displayed covariates as predictors of the primary endpoint response.

[b]Based on a reduced logistic model predicting the risk of primary endpoint response where covariates with a type III P-value >0.20 are not retained in the model via a backwards stepwise selection process.

[c]Sum of the number of times opioids were used on days where fentanyl ONLY was reported and counting 1 per day on days where fentanyl use was reported in addition to other opioids in the past 7 days at Screening (self-report). This is an estimate of the minimum number of times fentanyl was used per week at Baseline.

[d]"Injecting participants" was used as the reference group. Oral includes participants who checked "Other".

BUP-XR extended-release buprenorphine; CI, confidence interval; n, number of participants included in analysis; N, total participants; NA, not applicable; UDS, urine drug screen.

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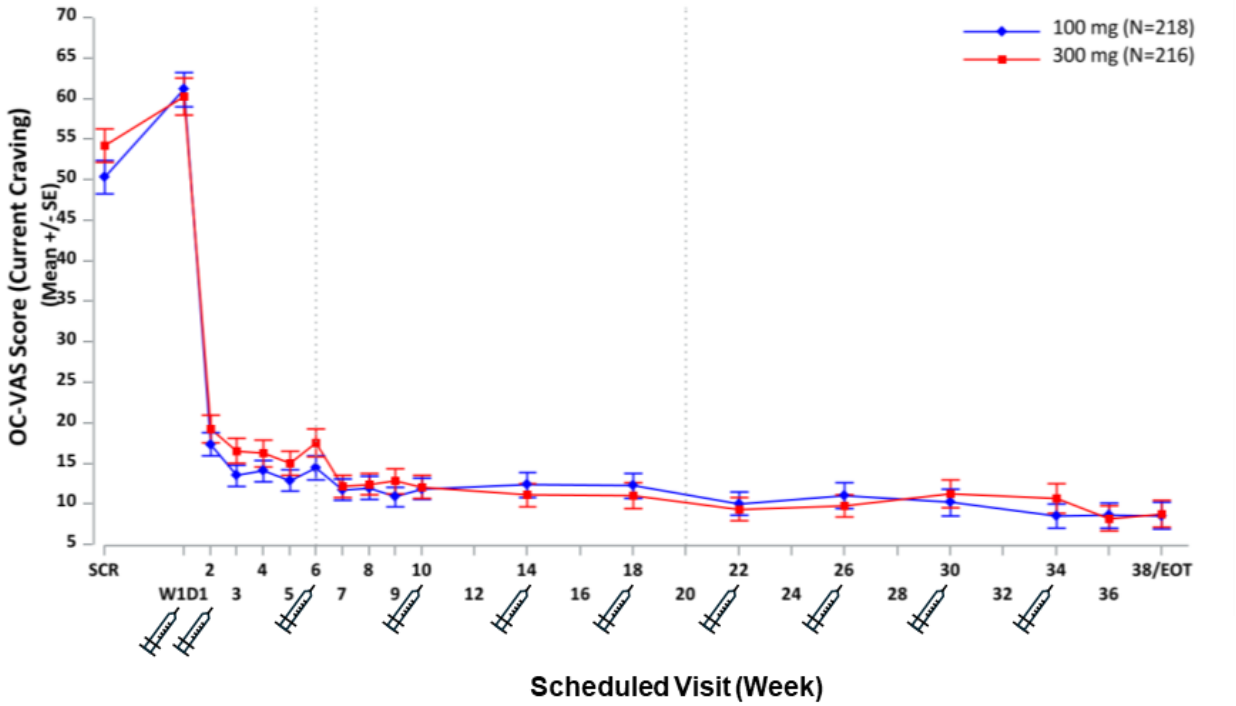
eFigures

eFigure 1. Craving Over Time

A. Current craving



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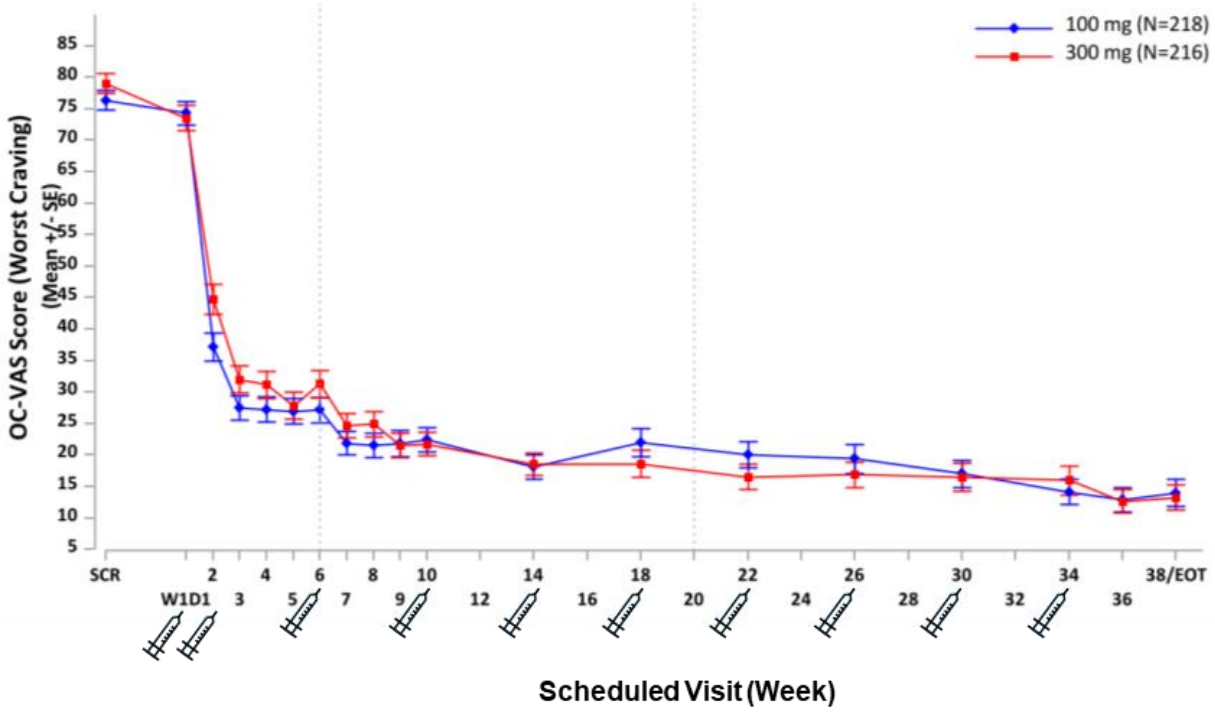
B. Worst craving last week



Shown for the efficacy population, defined as eligible participants who were randomized to double-blind treatment and received ≥1 BUP-XR maintenance injection and excluding 1 erroneously randomized participant who discontinued after BUP-XR injection 3 due to not meeting eligibility criteria. The X-axis numbers are the week # visit and syringe icons denote BUP-XR injection visits.
BUP-XR extended-release buprenorphine; EOT, end of treatment; OC-VAS, Opioid craving Visual Analog Scale; SCR, screening visit SE, standard error; W1D1, week 1 day 1.

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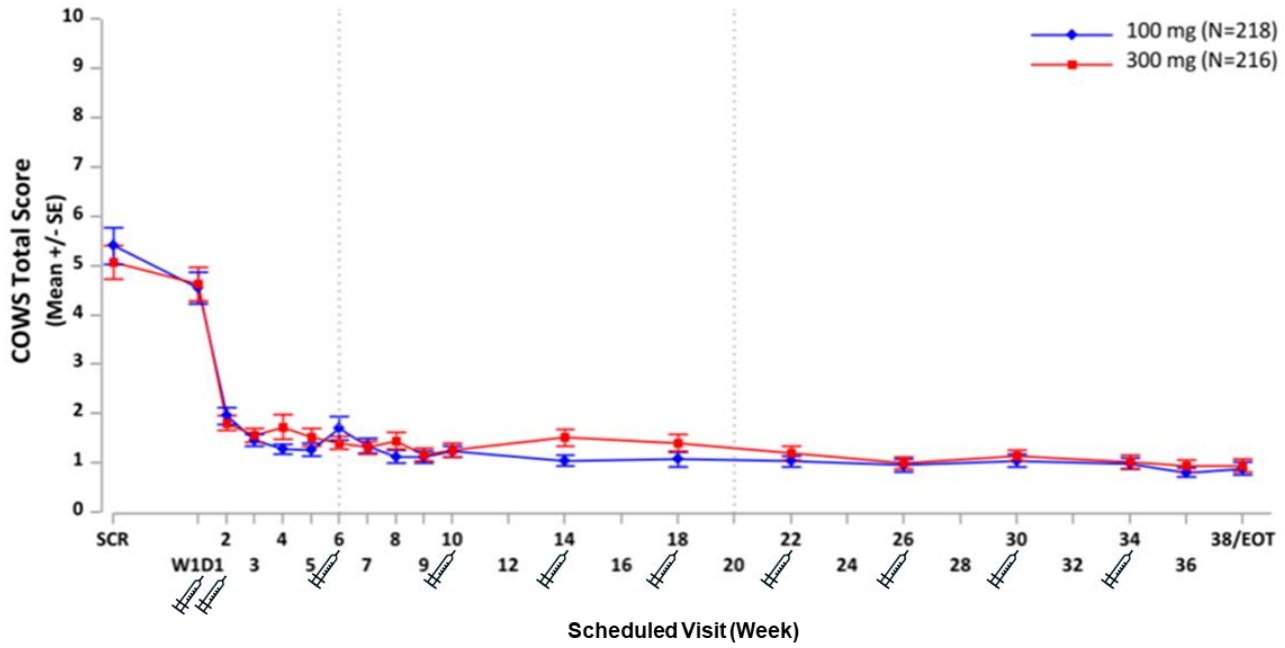
eFigure 2. Clinical Opiate Withdrawal Scale Over Time



Shown for the efficacy population, defined as eligible participants who were randomized to double-blind treatment and received ≥1 BUP-XR maintenance injection and excluding 1 erroneously randomized participant who discontinued after BUP-XR injection 3 due to not meeting eligibility criteria. The X-axis numbers are the week # visit and syringe icons denote BUP-XR injection visits.

Week 1 Day 1 values were taken as the first assessment post BUP-XR injection.

A COWS total score of 0 to 4 is defined as no active withdrawal, 5 to 12 is mild, 13 to 24 is moderate, 25 to 36 is moderately severe and >36 is severe withdrawal.

BUP-XR extended-release buprenorphine; COWS, Clinical Opiate Withdrawal Scale; EOT, end of treatment; SCR, screening visit; SE, standard error; W1D1, Week 1 Day 1.

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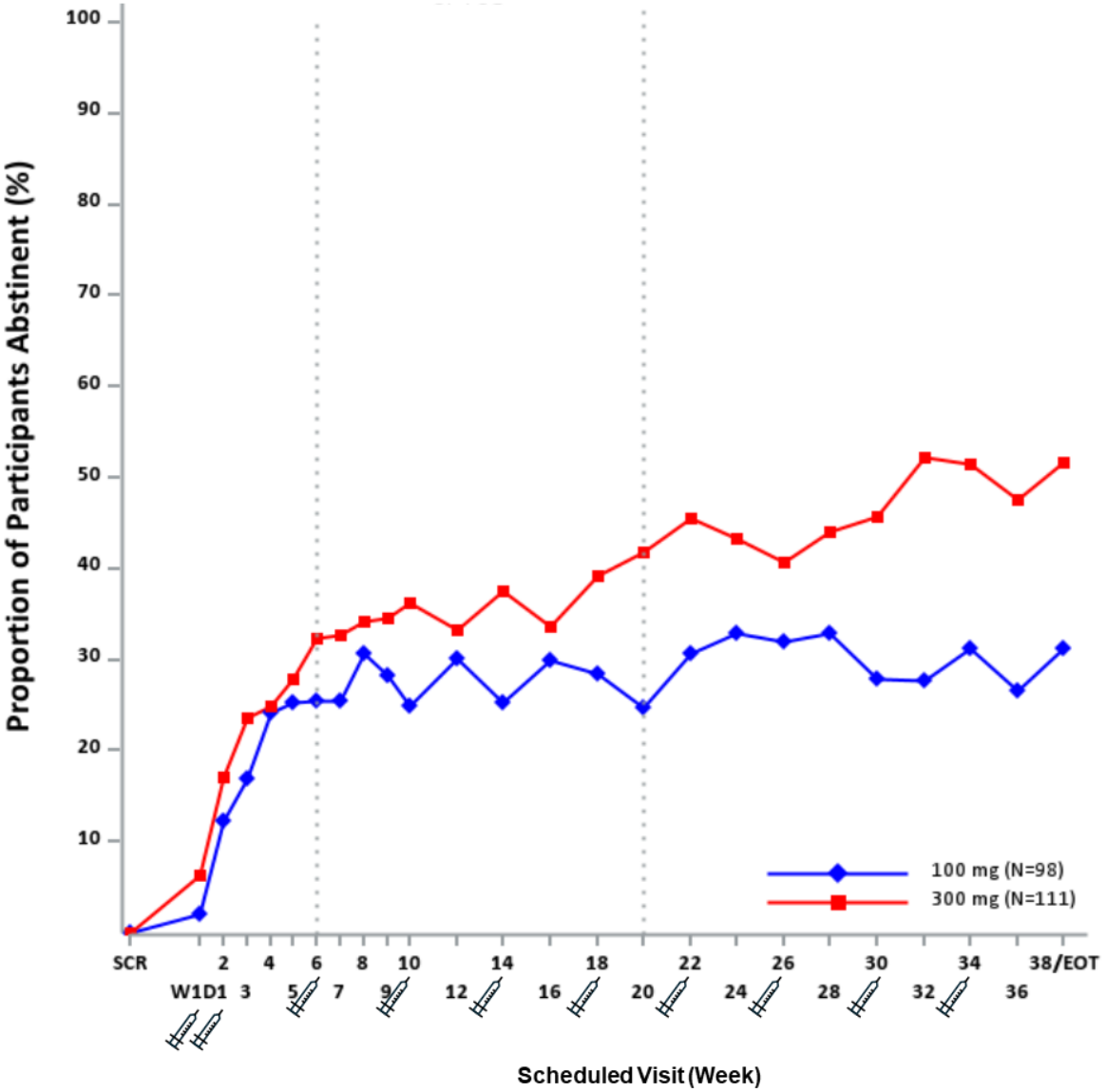
eFigure 3. Proportion of Participants With Opioid Abstinence by Visit for Heavy Fentanyl Users at Screening (as Observed)

A. Daily fentanyl use



Number at risk by visit

	SCR	W1D1	6	10	18	26	34	36	38/ EOT
100-mg	98	98	98	92	81	72	64	60	61
300-mg	111	111	111	102	84	76	68	63	62

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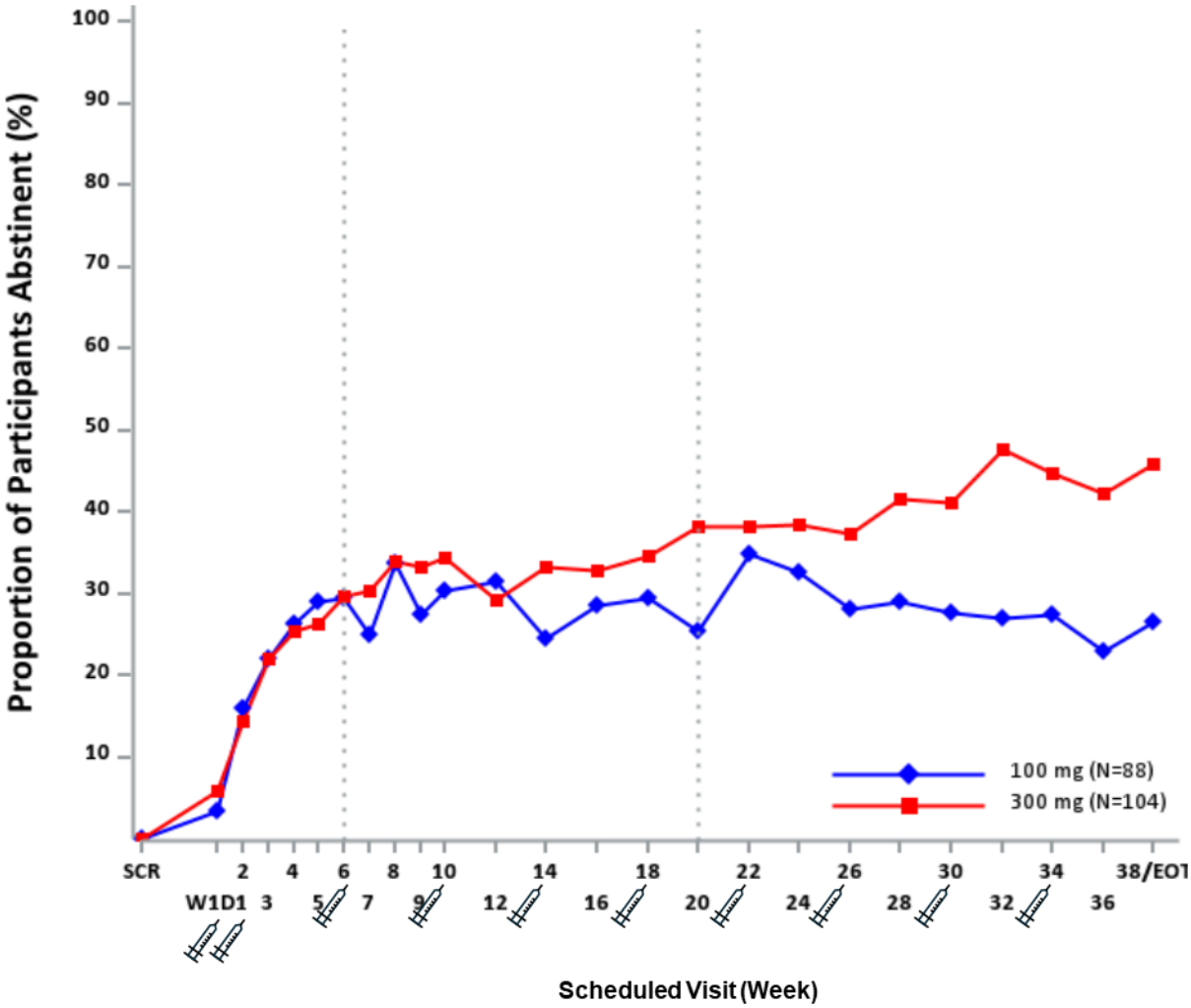
B. Fentanyl used ≥14 times/week[a]



Number at risk by visit

	SCR	W1D1	6	10	18	26	34	36	38/ EOT
100-mg	88	88	88	79	68	57	51	48	49
300-mg	104	104	104	96	81	75	67	64	61

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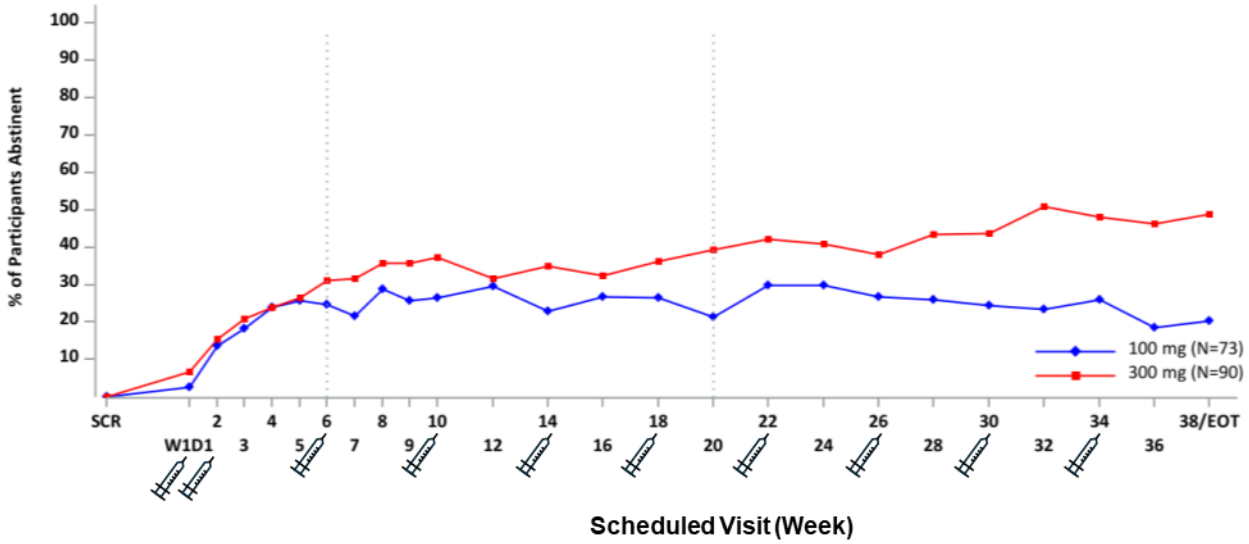
C. Fentanyl use both daily and ≥14 times/week[a]



Number at risk by visit

	SCR	W1D1	6	10	18	26	34	36	38/ EOT
100-mg	73	73	73	68	60	52	46	43	44
300-mg	90	90	90	83	69	63	56	54	51

Shown for the efficacy population, defined as eligible participants who were randomized to double-blind treatment and received ≥1 BUP-XR maintenance injection and excluding 1 erroneously randomized participant who discontinued after BUP-XR injection 3 due to not meeting eligibility criteria. The X-axis numbers are the week # visit and syringe icons denote BUP-XR injection visits. The proportion of participants abstinent for a given visit was calculated as the number of participants abstinent (by opioid negative UDS and self-report) divided by the number of participants with an observed opioid use assessment for that visit.

a. Sum the number of times opioids were used on days where fentanyl only was reported and counting 1 per day on days were fentanyl use was reported in addition to other opioids in the past 7 days at screening (self-report). This is an estimate of the minimum number of times fentanyl was used per week at baseline. At threshold of 14x/week was derived via a univariate logistic cut point analysis as the optimal value that would predict primary endpoint response.

BUP-XR extended-release buprenorphine; EOT, end of treatment; OC-VAS, Opioid craving visual analog scale; SCR, Screening visit SE, standard error; W1D1, week 1 day 1.

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